# HARIRAMAKRRISHNAN RAMACHANDRAN

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## PROFESSIONAL EXPERIENCE — 3.5 YEARS

**Software Engineer — HCL Technologies, Chennai, India** Sep 2021 – Jan 2025

- Designed and implemented scalable enforcement workflows using **Python** and **SQL** to address monetization quality challenges, analyzing high-volume product metadata.
- Performed deep-dive data analyses to identify system gaps and novel threat vectors within the YouTube advertising ecosystem, reducing policy violations by **18%**.
- Collaborated with Engineering and Trust & Safety partners to gather technical requirements, explain model architectures, and iterate on system improvements for anti-abuse protections.
- Wrote and optimized **complex SQL queries** and **Python scripts** to extract and analyze datasets, computing classifier performance metrics for applied AI systems.
- Reviewed controversial content to validate and improve the precision and recall of automated protection systems, contributing to a **15%** reduction in false positives.
- Translated complex analytical projects and technical concepts into actionable product insights for cross-functional and executive audiences, influencing long-term product decisions.
- Containerized Python-based analytical services using **Docker** and deployed them to a **Kubernetes** cluster on **AWS**, ensuring predictable rollout and resource management.

## SKILLS

**Languages** – Python, SQL, Java, R

**Cloud & Orchestration** – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes

**Data Analysis & ML** – Machine Learning, Generative AI, LLMs, scikit-learn, pandas, NumPy, Statistical Analysis, Data Visualization, Power BI, Tableau, DAX

**Databases** – PostgreSQL, MySQL, Oracle

**Development Tools & Methodologies** – Git, Jenkins, Jira, Confluence, Agile, REST APIs, FastAPI, Spring Boot

## PROJECTS

### YouTube Monetization Quality Analysis Platform

- Developed a **Python**-based data analysis pipeline to process product metadata, identifying patterns in monetization policy violations and feeding into enforcement workflows.
- Engineered **SQL queries** across large datasets to extract relevant features for machine learning models, improving classifier training accuracy by **12%**.
- Designed and implemented **dashboards in Power BI** to visualize abuse journeys and track key performance indicators for monetization quality, enabling faster stakeholder decision-making.

### Anti-Abuse ML Model for Content Moderation

- Trained a **scikit-learn** classifier on a dataset of ~150,000 content pieces to detect policy violations, achieving approximately **0.88 AUC** and **0.45 F1-score**.
- Developed a **Python** feature engineering script to extract signals related to content metadata and user behavior, directly contributing to the model's performance.
- Packaged the trained model behind a **FastAPI** inference endpoint, achieving a p95 latency of under **200ms** for real-time content analysis.

## EDUCATION

**M.Sc. Data Analytics — National College of Ireland, Dublin** Feb 2026

**B.E. Computer Science — SNS College of Technology, Coimbatore, India** 2021

## CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)